

$$G_{12}^k = \text{Diagram 1} = \text{Diagram 2} + \text{Diagram 3}$$

The diagrammatic equation for the two-point Green's function G_{12}^k is shown. The left-hand side is the full Green's function. The right-hand side is the sum of two diagrams:

- Diagram 1 (Left):** A horizontal line with an arrow pointing right. The left end is labeled '1' and the right end is labeled '2'. Below the arrow is the label k .
- Diagram 2 (Middle):** A horizontal line with an arrow pointing right. The left end is labeled '1' and the right end is labeled '2'. Below the arrow is the label k .
- Diagram 3 (Right):** A horizontal line with an arrow pointing right. The left end is labeled '1' and the right end is labeled '2'. Below the arrow is the label k . The line is divided into two segments by a circle. The segment to the left of the circle is labeled 'a' and the segment to the right is labeled 'b'. Inside the circle is the label Σ_{ab}^k .